# EXPERIMENT NO. 7

## Unique Elements from a List

Programming Language: Java

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## Aim

To write a Java program that creates a new list containing only the unique elements of the input list and to find its space complexity.

## Question

Write a program that takes an input list of n numbers and creates a new list containing only the unique elements from the original list. What is the space complexity of the algorithm?

**Test Case 1 - Some Duplicate Elements**

Input: [3, 7, 3, 5, 2, 5, 9, 2]

Expected Output: [3, 7, 5, 2, 9] (order may vary)

**Test Case 2 - Negative and Positive Numbers**

Input: [-1, 2, -1, 3, 2, -2]

Expected Output: [-1, 2, 3, -2] (order may vary)

**Test Case 3 - List with Large Numbers**

Input: [1000000, 999999, 1000000]

Expected Output: [1000000, 999999]

## Algorithm

Step 1: Start.

Step 2: Read the input list a of size n. Create an empty result list res and an empty HashSet seen.

Step 3: For each element x of a, try to add x to seen.

Step 4: If x was not already present in seen, append x to res.

Step 5: Print the list res.

Step 6: Stop.

## Program (Java)

```
import java.util.ArrayList;
import java.util.HashSet;
import java.util.List;
import java.util.Set;

public class Main {
```

```
    static List<Integer> unique(int[] a) {
        Set<Integer> seen = new HashSet<>();
        List<Integer> result = new ArrayList<>();
        for (int x : a) {
            if (seen.add(x))        // add() returns false if x is already present
                result.add(x);
        }
        return result;
    }

    public static void main(String[] args) {
        int[] t1 = {3, 7, 3, 5, 2, 5, 9, 2};
        int[] t2 = {-1, 2, -1, 3, 2, -2};
        int[] t3 = {1000000, 999999, 1000000};

        System.out.println("Output: " + unique(t1));
        System.out.println("Output: " + unique(t2));
        System.out.println("Output: " + unique(t3));
    }
}
```

## Output

```
Output: [3, 7, 5, 2, 9]
Output: [-1, 2, 3, -2]
Output: [1000000, 999999]

=== Code Execution Successful ===
```

## Complexity Analysis

**Time Complexity:** O(n) on average - each element is checked and inserted in the HashSet in O(1) expected time (O(n^2) if a plain list search is used instead).

**Space Complexity:** O(n) - the HashSet and the result list can each hold up to n elements when all elements are distinct.

## Result

Thus the program was executed successfully and the output was verified.